

The robustness valley of the Cauchy combination test: exact calibration at both independence and comonotonicity, worst in between

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Abstract

The Cauchy combination test (ACAT) of Liu & Xie merges p -values through a tangent transform and reads its p -value off a standard-Cauchy reference. Its appeal is a reference law that is *exactly* correct under independence and *approximately* correct — in the far tail — under arbitrary dependence, which is why it is now ubiquitous in genomics and large-scale testing. This note makes two points. First, the independence-exactness is best understood not as a lucky identity but as **1-stability**: the standard symmetric Cauchy is the unique strictly stable law closed under *all* convex combinations without rescaling, so the ACAT reference is invariant to the number of p -values and to the weights, with no asymptotics anywhere. Second, and less obviously, exactness returns at the *other* extreme of dependence: under a one-factor equicorrelated Gaussian copula with correlation ρ , the null statistic is exactly standard Cauchy at both $\rho = 0$ (independence) **and** $\rho = 1$ (comonotonicity), for every finite m . The fixed-level size distortion is therefore a **non-monotone** function of ρ — zero at both endpoints, strictly positive in the interior — a “robustness valley.” Simulation locates the interior worst case and shows the inflation grows with m (up to $1.7\times$ at $\alpha = 0.05$, $m = 100$) while the worst-case correlation drifts toward 0, so the comonotone exactness thins into a narrow boundary feature as m grows. The practical reading: ACAT is safe when p -values are nearly independent *or* nearly comonotone; moderate correlation, not strong correlation, is where its finite-sample calibration is worst.

1. Setup and notation

Given input p -values $p_1, \dots, p_m$ and nonnegative weights $w_1, \dots, w_m$ with $\sum_i w_i = 1$, the Cauchy combination (ACAT) statistic is

$$T = \sum_{i=1}^m w_i \tan(\pi(\tfrac{1}{2} - p_i)) = \sum_{i=1}^m w_i \cot(\pi p_i), \quad (1)$$

using $\tan(\pi/2 - x) = \cot(x)$. The reported (one-sided) p -value is $P_{\text{ACAT}} = \frac{1}{2} - \arctan(T)/\pi$, and a level- α test rejects when $P_{\text{ACAT}} \leq \alpha$, i.e. when

$$T \geq c_\alpha := \cot(\pi\alpha), \quad (2)$$

the upper- α quantile of the standard Cauchy law. Throughout we study the **global null**, under which each $p_i \sim U(0, 1)$ marginally, and we take the canonical dependence model used in the ACAT literature: a **one-factor equicorrelated Gaussian copula**. Latent scores

$$Z_i = \sqrt{\rho} F + \sqrt{1-\rho} \varepsilon_i, \quad F, \varepsilon_1, \dots, \varepsilon_m \stackrel{\text{iid}}{\sim} N(0, 1), \quad (3)$$

give equicorrelated standard normals ($\text{Corr}(Z_i, Z_j) = \rho$ for $i \neq j$) and one-sided p -values $p_i = 1 - \Phi(Z_i)$, each $U(0, 1)$. Write $C_i := \cot(\pi p_i)$ for the transformed scores, so $T = \sum_i w_i C_i$.

The one basic fact we use repeatedly: if $U \sim U(0, 1)$ then $\cot(\pi U)$ has the **standard Cauchy** distribution (its cdf is $\frac{1}{2} + \arctan(x)/\pi$, checked by inverting), with characteristic function $\mathbb{E} e^{itC} = e^{-|t|}$.

2. Why Cauchy: combination-invariance is 1-stability

The design of (1) is usually justified by “a weighted sum of standard Cauchys is standard Cauchy.” It is worth stating exactly *which* property of the Cauchy law is doing the work, because it pins down the reference distribution completely and explains the invariances practitioners rely on.

Recall that a random variable X is **strictly α -stable** ($0 < \alpha \leq 2$) if for iid copies $X_1, \dots, X_n$, $X_1 + \dots + X_n \stackrel{d}{=} n^{1/\alpha} X$. More generally, for a symmetric strictly α -stable X and constants a_i , $\sum_i a_i X_i \stackrel{d}{=} (\sum_i |a_i|^\alpha)^{1/\alpha} X$.

Proposition 1 (combination-invariance forces $\alpha = 1$). *Let X be a nondegenerate symmetric strictly α -stable random variable. The convex-combination map $(X_1, \dots, X_m) \mapsto \sum_i w_i X_i$ (iid copies, $w_i \geq 0$, $\sum_i w_i = 1$) preserves the law of X for every m and every such weight vector if and only if $\alpha = 1$. The symmetric strictly 1-stable laws are exactly the centered Cauchy family; normalizing the transform in (1) to the standard Cauchy fixes the scale.*

Proof. For symmetric strictly α -stable X , $\sum_i w_i X_i \stackrel{d}{=} (\sum_i w_i^\alpha)^{1/\alpha} X$. Preservation for all convex w requires $(\sum_i w_i^\alpha)^{1/\alpha} = 1$ whenever $\sum_i w_i = 1$. If $\alpha = 1$ this is the identity $\sum_i w_i = 1$, so it holds for all m, w . If $\alpha \neq 1$, take $m = 2$, $w = (t, 1-t)$: the scale is $g(t) = (t^\alpha + (1-t)^\alpha)^{1/\alpha}$, and $g(\frac{1}{2}) = 2^{1/\alpha-1} \neq 1$ for $\alpha \neq 1$, so the law is not preserved. Symmetric 1-stable laws are the centered Cauchys (Samorodnitsky & Taqqu, Ch. 1); the tangent transform sends $U(0, 1)$ to the *standard* Cauchy, fixing the scale to 1. ■

This is the crisp “why Cauchy.” Gaussians are 2-stable, so averaging m of them shrinks the scale by $\sqrt{m}$ — a moving reference. Only at $\alpha = 1$ does the *simple* average leave the law untouched, and among 1-stable laws symmetry plus the closed-form quantile of (2) select the Cauchy. It also makes the two invariances that ACAT users take for granted into one statement: because the reference is a *fixed point* of convex combination, it does not depend on m and does not depend on the weights. The extension to other α -stable transforms (Ling & Rho, 2021) is exactly the observation that if you are willing to rescale by $(\sum_i w_i^\alpha)^{1/\alpha}$ you may use any stable law; ACAT is the rescaling-free special case.

An immediate consequence is the exact independence null, with no appeal to asymptotics.

Corollary 1 (exactness under independence). *If $p_1, \dots, p_m$ are independent $U(0, 1)$, then T in (1) is exactly standard Cauchy for every m and every convex weight vector.*

Proof. $C_i = \cot(\pi p_i)$ are iid standard Cauchy; by independence $\mathbb{E} e^{itT} = \prod_i e^{-|w_i t|} = e^{-|t| \sum_i w_i} = e^{-|t|}$. ■

3. A symmetry that survives all dependence

Before the endpoints, one exact fact holds for *every* ρ in the model (3) and is worth isolating, because it tells us the distortion is a pure tail phenomenon.

Proposition 2 (symmetry). *Under the one-factor model (3) with any $\rho \in [0, 1]$ and any weights, $T \stackrel{d}{=} -T$; in particular the null median of T is 0.*

Proof. The map $(F, \varepsilon) \mapsto (-F, -\varepsilon)$ preserves the law of the Gaussian noise and sends $Z_i \mapsto -Z_i$, hence $p_i = 1 - \Phi(Z_i) \mapsto \Phi(Z_i) = 1 - p_i$ and $C_i = \cot(\pi p_i) \mapsto \cot(\pi(1 - p_i)) = -\cot(\pi p_i) = -C_i$. Thus $T \mapsto -T$ while the distribution of the inputs is unchanged, giving $T \stackrel{d}{=} -T$. ■

So dependence cannot shift ACAT’s center; whatever miscalibration exists is a change in the *scale/tail* of a symmetric law, not a location bias. The simulation confirms the median sits at 0 to Monte-Carlo precision at $\rho = 0, 0.5, 1$ alike.

4. Exactness returns at comonotonicity

The nontrivial endpoint is $\rho = 1$.

Proposition 3 (exactness under comonotonicity). *In model (3) with $\rho = 1$, T is exactly standard Cauchy for every m and every convex weight vector.*

Proof. At $\rho = 1$ the idiosyncratic term vanishes: $Z_i = F$ for all i , so all p_i equal the single $U(0, 1)$ variate $p := 1 - \Phi(F)$, and all $C_i = \cot(\pi p) =: C$ coincide. Then $T = \sum_i w_i C = C \sum_i w_i = C$, a single standard Cauchy. ■

The mechanism is different from Corollary 1 — there the average of m *independent* Cauchys is Cauchy by 1-stability; here the “ m Cauchys” have collapsed to *one*. But the reference law that ACAT assumes is exactly correct at both ends. Between them it is not: for $0 < \rho < 1$ the conditional law of C_i given $F = f$ is neither standard Cauchy (that is $\rho = 0$) nor degenerate (that is $\rho = 1$). Conditionally on $F = f$ the scores $Z_i \sim N(\sqrt{\rho} f, 1 - \rho)$ are independent but no longer standard normal, so C_i has a Cauchy-*type* tail with a conditional scale and a location that both depend on f (this conditional Cauchy-tail structure is the content of Liu & Xie’s dependence theorem). The unconditional law of T is then a genuine scale–location *mixture* of approximately-Cauchy laws over $f \sim N(0, 1)$, and such a mixture equals a single standard Cauchy only in the two degenerate limits where the mixing disappears (no spread at $\rho = 0$; a single atom that integrates back to Cauchy at $\rho = 1$). Hence:

Claim (the valley). *The fixed-level size $S_m(\rho, \alpha) := \mathbb{P}_\rho(T \geq c_\alpha)$ satisfies $S_m(0, \alpha) = S_m(1, \alpha) = \alpha$ exactly (Corollary 1, Proposition 3), and is $> \alpha$ for $0 < \rho < 1$, with an interior maximum. The two endpoint equalities are theorems; the strict interior inflation and the location of its maximum we establish by simulation (§5) — a fully rigorous interior bound is left open (§6).*

5. Simulation: locating the valley

All numbers below are from `sim.py` (seed 20260713, one-factor model (3), equal weights $w_i = 1/m$, 2×10^6 Monte-Carlo draws per (ρ, m) cell and 4×10^6 at the endpoints). Every reported figure is the code's actual output.

Endpoints are exact. At $\rho = 0$ and $\rho = 1$ the empirical size matches the nominal level to Monte-Carlo error for all m and all α tested:

m	endpoint	$\alpha=0.05$	$\alpha=0.01$	$\alpha=0.001$
5	$\rho = 0$	0.04998	0.00995	0.00098
5	$\rho = 1$	0.05012	0.00995	0.00102
20	$\rho = 0$	0.04999	0.00995	0.00098
20	$\rho = 1$	0.05007	0.01006	0.00099

(Monte-Carlo SE $\approx 1.1 \times 10^{-4}$ at $\alpha = 0.05$.) A Kolmogorov–Smirnov test of the full statistic against the standard Cauchy does not reject at either endpoint ($m = 5$: $D = 0.0029$, $p = 0.07$ at $\rho = 0$; $D = 0.0015$, $p = 0.77$ at $\rho = 1$), and the null median is 0 to three decimals at $\rho = 0, 0.5, 1$ — consistent with Proposition 2.

The interior inflates, and the endpoints do not. Sweeping ρ produces the valley of Figure 1: the size dips to α at both ends and bulges above it in between.

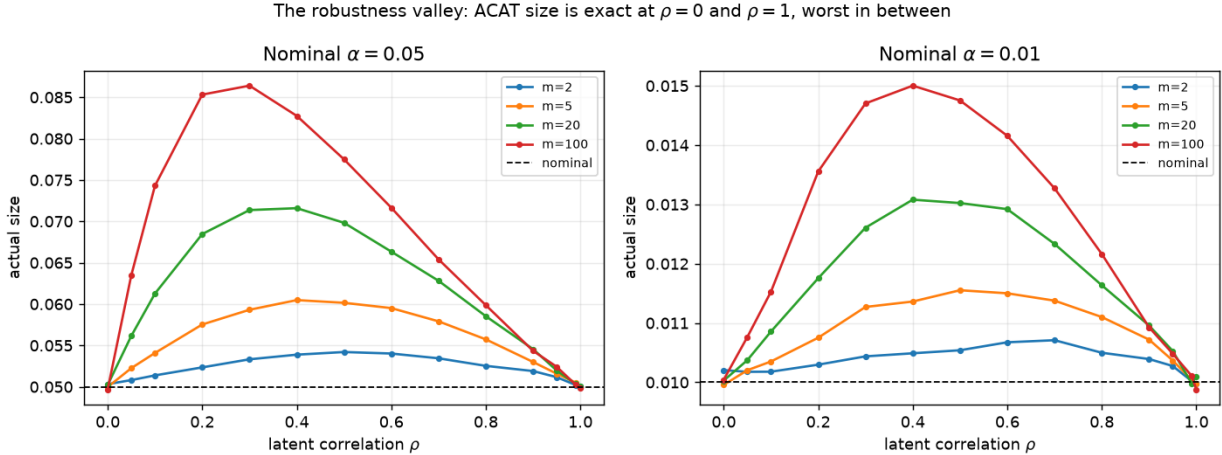


Figure 1: Actual size of ACAT versus latent correlation ρ , at nominal $\alpha = 0.05$ (left) and $\alpha = 0.01$ (right), for $m \in \{2, 5, 20, 100\}$. The dashed line is the nominal level. Size is exact at $\rho = 0$ and $\rho = 1$ and inflated in the interior; the bulge deepens with m and its worst case drifts toward smaller ρ .

The worst-case inflation and its location:

m	α	size at ρ^*	ρ^*	inflation
2	0.05	0.0542	0.50	$1.08\times$
5	0.05	0.0605	0.40	$1.21\times$
20	0.05	0.0716	0.40	$1.43\times$

m	α	size at ρ^*	ρ^*	inflation
100	0.05	0.0864	0.30	1.73×
20	0.01	0.0131	0.40	1.31×
100	0.01	0.0150	0.40	1.50×

Three features stand out. (i) The inflation grows with m : at $\alpha = 0.05$ it climbs from $1.08\times$ ($m = 2$) to $1.73\times$ ($m = 100$). (ii) The worst-case correlation ρ^* drifts *toward* 0 as m grows ($0.50 \rightarrow 0.40 \rightarrow 0.30$ at $\alpha = 0.05$); the comonotone endpoint stays exact but its neighborhood of near-exactness narrows, so for large m the exact-at- $\rho = 1$ result is a thin boundary feature rather than a wide safe region. This is the finite- m shadow of the large- m phenomenon studied by the boundary-layer calibration work of 2026 (arXiv:2603.22668): at *fixed* positive ρ the statistic ceases to have any universal fixed-level reference as $m \rightarrow \infty$; our valley shows that for finite m the two boundaries $\rho \in \{0, 1\}$ are the exact exceptions. (iii) The valley is shallower at smaller α — at $m = 100$ the peak inflation falls from $1.73\times$ ($\alpha = 0.05$) to $1.50\times$ ($\alpha = 0.01$) — the finite-sample echo of Liu & Xie’s tail robustness, which guarantees the ratio $\rightarrow 1$ as $\alpha \rightarrow 0$.

Figure 2 shows the same story on the tail directly: the empirical upper tail $\mathbb{P}(T > c)$ sits exactly on the standard-Cauchy reference at $\rho \in \{0, 1\}$ and lies visibly above it at $\rho = 0.5, 0.9$.

6. Discussion

What this changes. The usual mental model of ACAT is “exact under independence, degrades as dependence increases.” The valley corrects the second half: degradation is not monotone in dependence. Strong positive dependence is *not* the adversarial case at the level of a single equicorrelated block — comonotone inputs are handled exactly, because the combination collapses to one p -value. The genuinely hard regime is *moderate* correlation, where the statistic is a nontrivial mixture of conditional-Cauchy laws. For a practitioner this reframes the safety check: near-independent and near-duplicated p -values are both fine; a block of, say, $\rho \approx 0.3$ – 0.5 tests is where a nominal 0.05 can quietly become 0.07–0.09.

Relation to known results. The independence-exactness and the tail robustness are Liu & Xie (2020); we have only reorganized the first as 1-stability (Proposition 1) to explain the m - and weight-invariance, and used the second to explain why the valley shrinks with α . The comonotone exactness (Proposition 3) and the resulting non-monotonicity of the fixed-level size in ρ are, to our reading of the literature (including the stable-combination extension of Ling & Rho and the 2026 boundary-layer calibration), the new observations here. They sit beside, not against, the large- m message that fixed positive correlation destroys the universal reference: both are true, on different axes (ρ at finite m versus m at fixed ρ).

Limitations. (i) The interior claim $S_m(\rho, \alpha) > \alpha$ and the existence of a single interior maximum are established by simulation; we prove only the two endpoint equalities and the exact symmetry. A rigorous interior lower bound — e.g. showing the conditional-Cauchy mixture is tail-heavier than standard Cauchy for all $0 < \rho < 1$ — would upgrade the Claim to a theorem, and a second-order expansion of S_m near $\rho = 0$ would pin the initial rate of inflation. (ii) We treat a *single* equicorrelated block with the canonical one-factor Gaussian copula and one-sided p -values; general covariance, two-sided inputs, mixed-sign correlations, and multi-block structure are not covered, and negative dependence (outside $[0, 1]$ here) may behave differently. (iii) The worst-case inflations reported ($\leq 1.7\times$ over the grid) are for the null only; we say nothing about the power cost of any

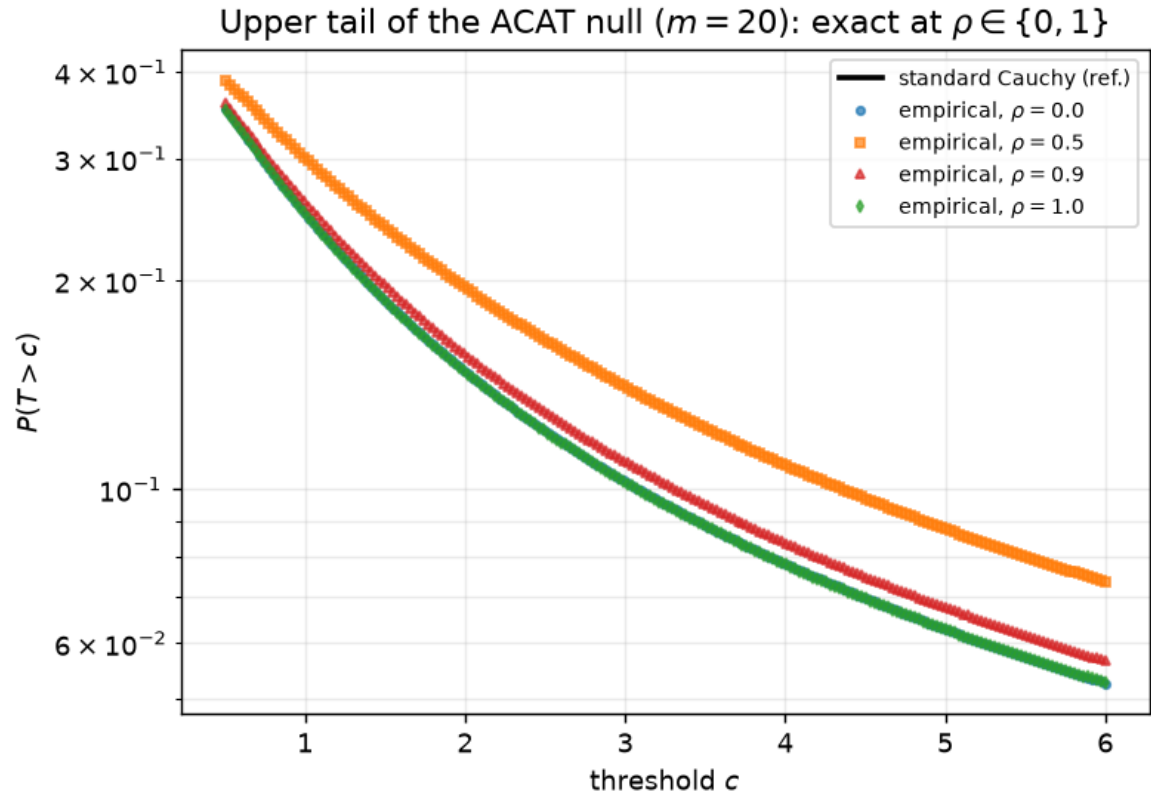


Figure 2: Upper tail of the ACAT null for $m = 20$, log scale. Empirical tails at $\rho = 0$ and $\rho = 1$ track the standard-Cauchy reference; intermediate correlations $\rho = 0.5, 0.9$ have heavier tails (anti-conservative), while both endpoints are exact.

recalibration. The boundary-layer reference of arXiv:2603.22668 is the natural fix for the large- m interior; our contribution is diagnostic — knowing *where* on the ρ axis the finite- m distortion actually lives, and that it is exactly zero at both ends.

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